IMEN891K / IDSC501 / AXGS-501 · DATA ENGINEERING · FALL 2026

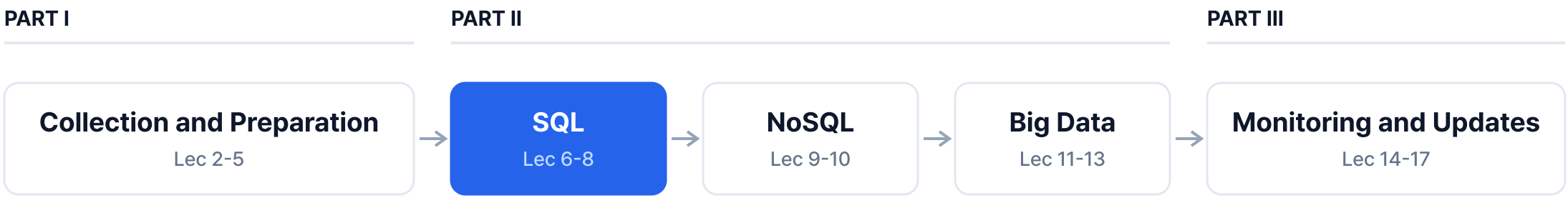
6.

SQL I: Basics

Sangwoo Mo · POSTECH

Industrial & Management Engineering

Where We Are


PART I
PART II
PART III
Collection and Preparation
Lec 2-5
SQL
Lec 6-8
NoSQL
Lec 9-10
Big Data
Lec 11-13
Monitoring and Updates
Lec 14-17

Outline: SQL I

01	**Read a Table**	find the rows and columns you need
02	**Refine a Result**	remove repeats, sort, and keep top rows
03	**Handle Missing Values**	handle rows with absent values
04	**Write Safely**	change data while keeping tables consistent

Tables and SQL

A database stores related tables; SQL reads and changes their rows

Term	Today
Table	Named rows and columns
Schema	Column names, types and constraints
SQL	The language for reading and writing tables
SQLite	The database engine running inside this deck

Schema

- *Logical Schema*
  - Students(sid: *string*, name: *string*, gpa: *float*)
  - Courses(cid: *string*, cname: *string*, credits: *int*)
  - Enrolled(sid: *string*, cid: *string*, grade: *string*)

Students

sid	Name	Gpa
101	Bob	3.2
123	Mary	3.8

Relations

Enrolled

sid	cid	Grade
123	564	A

Courses

cid	cname	credits
564	564-2	4
308	417	2

Case study: A Product Catalogue

```
Product(pname, price, category, manufacturer)
```

pname	price	category	manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	29.99	Gadgets	GizmoWorks
SingleTouch	149.99	Photography	Canon
MultiTouch	203.99	Household	Hitachi

Find products by category, compare prices and prepare a short price list

Four products form the input for filtering, choosing columns and sorting prices

SQL Query

- Basic form (there are many many more bells and whistles)

```
SELECT <attributes>
FROM   <one or more relations>
WHERE  <conditions>
```

Call this a SFW query.

SQL Query: Selection

Selection is the operation of filtering a relation's tuples on some condition

```
SELECT *
FROM   Product
WHERE  Category = 'Gadgets'
```

selection: keep rows

PName	Price	Category	Manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	29.99	Gadgets	GizmoWorks
SingleTouch	149.99	Photography	Canon
MultiTouch	203.99	Household	Hitachi



PName	Price	Category	Manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	29.99	Gadgets	GizmoWorks

SQL Query: Projection

Projection is the operation of producing an output table with tuples that have a subset of their prior attributes

```
SELECT Pname, Price,
Manufacturer
```

projection: keep columns

```
FROM   Product
WHERE  Category = 'Gadgets'
```

PName	Price	Category	Manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	29.99	Gadgets	GizmoWorks
SingleTouch	149.99	Photography	Canon
MultiTouch	203.99	Household	Hitachi



PName	Price	Manufacturer
Gizmo	19.99	GizmoWorks
PowerGizmo	29.99	GizmoWorks

A Query Produces a Table

Input schema

Product(PName, Price, Category, Manufacturer)

```
SELECT Pname, Price, Manufacturer
FROM   Product
WHERE  Category = 'Gadgets'
```



Output schema

Answer(PName, Price, Manufacturer)

Strings and Conditions

```sql
SELECT pname, price
FROM Product
WHERE category = 'Gadgets' AND price > 20;
```

↑ strings in single quotes

Parentheses Change the Filter

```
SELECT pname,category,price FROM Product
WHERE (category='Gadgets' OR category='Photography')
  ↑ OR inside the parentheses first, then AND
  AND price>20;
```

pname	category	price
PowerGizmo	Gadgets	29.99
SingleTouch	Photography	149.99

Parentheses apply the price condition to both selected categories

IN Lists the Allowed Values

```
SELECT pname,category FROM Product
WHERE category IN ('Gadgets','Photography');
```

↑ any value in the list

pname	category
Gizmo	Gadgets
PowerGizmo	Gadgets
SingleTouch	Photography

IN selects the three products in either named category

Recap: The Product Table

pname	price	category	manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	29.99	Gadgets	GizmoWorks
SingleTouch	149.99	Photography	Canon
MultiTouch	203.99	Household	Hitachi

Predict the remaining rows and output columns before writing the query

Demo: Read the Product Table

1. Return the name and price of every Gadget product
2. Keep only Gadget products priced above 20

```
-- write your query here, then Run
```

QUERY RESULT

Write a query on the left, then Run.

Answer: Read the Product Table

```
SELECT pname,price FROM Product
WHERE category='Gadgets' AND price>20;
```

pname	price
PowerGizmo	29.99

Adding the price condition keeps PowerGizmo and removes Gizmo

LIKE: Simple String Pattern Matching

```
SELECT *
FROM   Product
WHERE  PName LIKE '%gizmo%'
```

% = any run of characters

- s **LIKE** p: pattern matching on strings
- p may contain two special symbols:
  - % = any sequence of characters
  - _ = any single character

Dialect: LIKE is case-sensitive in PostgreSQL and case-insensitive for ASCII in SQLite (`'%gizmo%'` also matches Gizmo here).

Caveat: Duplicates in the Output

Projection keeps every copy; DISTINCT keeps one

```
SELECT DISTINCT Category
FROM   Product
```

drop duplicate rows



Category
Gadgets
Photography
Household

Versus

```
SELECT Category
FROM   Product
```



Category
Gadgets
Gadgets
Photography
Household

ORDER BY: Sorting the Results

```
SELECT   PName, Price, Manufacturer
FROM     Product
WHERE    Category='Gadgets' AND Price > 20
ORDER BY Price, PName
```

by Price, ties by PName

Ties are broken by the second attribute on the ORDER BY list, etc.

Ordering is ascending, unless you specify the DESC keyword.

LIMIT and Deterministic Order

```
SELECT pname, price FROM Product
ORDER BY price DESC, pname ASC LIMIT 2;
```

↑ first 2 rows after the sort

pname	price
MultiTouch	203.99
SingleTouch	149.99

Sort before limiting; an explicit tie-breaker makes the chosen rows reproducible

Recap: Categories and Prices

pname	price	category
Gizmo	19.99	Gadgets
PowerGizmo	29.99	Gadgets
SingleTouch	149.99	Photography
MultiTouch	203.99	Household

The input repeats Gadgets twice; the two Gadget prices are different

Demo: Prepare a Price List

1. List Gadget names and prices, highest price first; return only the first product
2. List each category once with `DISTINCT`

```
-- write your query here, then Run
```

QUERY RESULT

Write a query on the left, then Run.

Answer: Prepare a Price List

```sql
SELECT pname,price FROM Product
WHERE category='Gadgets'
ORDER BY price DESC,pname LIMIT 1;
```

pname	price
PowerGizmo	29.99

ORDER BY defines the order; LIMIT keeps the first row

Answer: Unique Categories

```sql
SELECT DISTINCT category FROM Product;
```

category
Gadgets
Photography
Household

Two Gadgets inputs become one category row

NULL and Missing Values

pname	price
Gizmo	19.99
PowerGizmo	NULL
SingleTouch	149.99
MultiTouch	203.99

NULL means the price is missing; test it with IS NULL

This copy has a missing PowerGizmo price; NULL is different from a zero price

NULL Comparisons

Expression	Result
NULL = NULL	UNKNOWN
NULL <> 5	UNKNOWN
NULL IS NULL	TRUE
NULL IS NOT NULL	FALSE

WHERE keeps only TRUE; FALSE and UNKNOWN both drop the row

An absent value needs IS NULL; equality cannot identify a missing match

Caveat: Missing Prices in a Filter

```
SELECT pname,price FROM Product WHERE price>20;
```

↑ a NULL price is not > 20: row dropped

pname	price
SingleTouch	149.99
MultiTouch	203.99

A missing price does not satisfy price greater than 20

Recap: The Missing-Price Copy

pname	price	category	manufacturer
Gizmo	19.99	Gadgets	GizmoWorks
PowerGizmo	NULL	Gadgets	GizmoWorks
SingleTouch	149.99	Photography	Canon
MultiTouch	203.99	Household	Hitachi

PowerGizmo is the only product with a missing price in this copy

Demo: Handle Missing Prices

1. Find products with a missing price
2. Return products priced above 20, and also include products with a missing price

```
-- write your query here, then Run
```

QUERY RESULT

Write a query on the left, then Run.

Answer: Handle Missing Prices

```
SELECT pname,price FROM Product
WHERE price>20 OR price IS NULL;
```

pname	price
PowerGizmo	NULL
SingleTouch	149.99
MultiTouch	203.99

The explicit NULL condition includes PowerGizmo alongside the two known prices above 20

Keys and References

cname	stockprice	country
GizmoWorks	25.0	USA
Canon	65.0	Japan
Hitachi	15.0	Japan

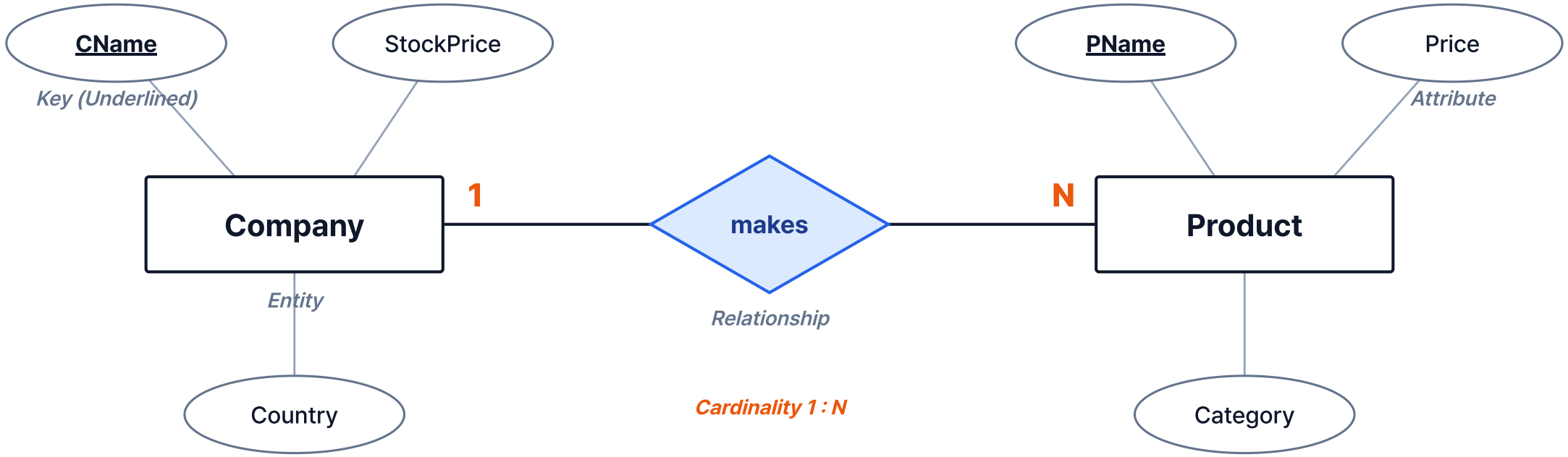
pname	manufacturer
Gizmo	GizmoWorks
PowerGizmo	GizmoWorks
SingleTouch	Canon
MultiTouch	Hitachi

A primary key identifies a row; a foreign key requires an existing referenced key

Each product names one existing company; the key columns identify the referenced rows

Entity–Relationship (ER) Diagram

An ER diagram describes entities, attributes and relationships before table design



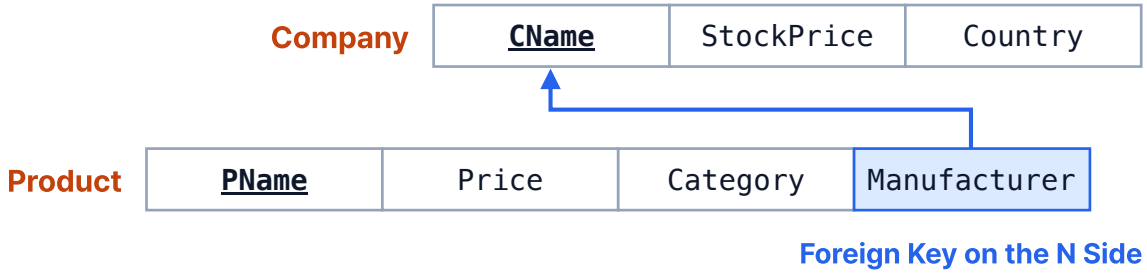
One company **makes** many products: **1** on one side, **N** on the other

ER to Tables

Each relationship becomes either a foreign key column or a table of its own

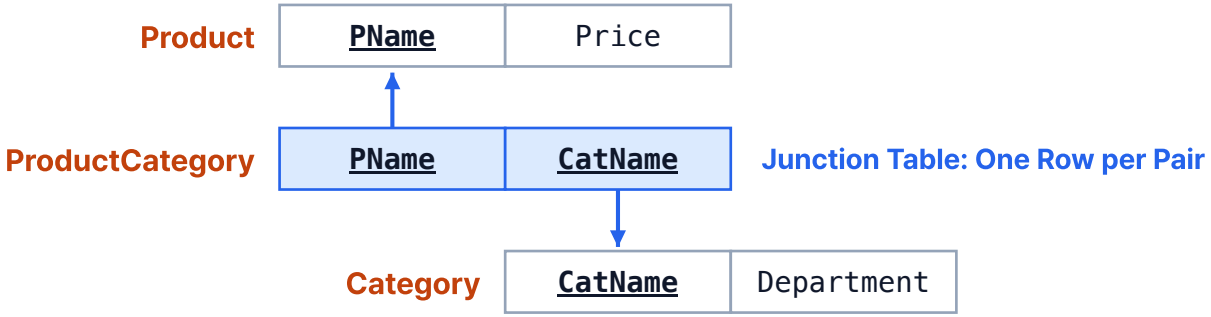
1 : N





N : M





Use a **foreign key** for 1:N and a **junction table** for N:M

Junction Table

If a product may sit in several categories, Category leaves Product and gets its own table

Product

PName	Manufacturer
Gizmo	GizmoWorks
PowerGizmo	GizmoWorks
SingleTouch	Canon
MultiTouch	Hitachi

ProductCategory

PName	CatName
Gizmo	Gadgets
PowerGizmo	Gadgets
SingleTouch	Photography
SingleTouch	Gadgets
MultiTouch	Household

Category

CatName	Department
Gadgets	Electronics
Photography	Electronics
Household	Home

```
ProductCategory: PRIMARY KEY (PName, CatName) · PName → Product.PName · CatName → Category.CatName
```

SingleTouch has two **category links** but one product row

A Composite Key Identifies a Pair

```
CREATE TABLE ProductCategory (
  pname TEXT NOT NULL REFERENCES Product(pname),
  catName TEXT NOT NULL,
  PRIMARY KEY (pname,catName)
);
```

the pair is unique

CREATE TABLE and INSERT

```
CREATE TABLE Product (
  pname TEXT PRIMARY KEY NOT NULL,
  price REAL, category TEXT, manufacturer TEXT NOT NULL,
  FOREIGN KEY (manufacturer) REFERENCES Company(cname)
    ↑ must exist in Company
);
INSERT INTO Product VALUES ('Gizmo',19.99,'Gadgets','GizmoWorks');
```

Caveat: A Repeated Primary Key

```
INSERT INTO Product VALUES
('Gizmo',99,'Gadgets','GizmoWorks');
```

SQLITE SAYS

```
UNIQUE constraint failed:
Product.pname
```

A repeated product key fails even when the other values differ

UPDATE and DELETE

```
UPDATE Product SET price = 24.99 WHERE
pname = 'Gizmo';
```

↑ WHERE picks the row to change

```
SELECT pname, price FROM Product;
```

pname	price
Gizmo	24.99
PowerGizmo	29.99
SingleTouch	149.99
MultiTouch	203.99

```
DELETE FROM Product WHERE pname = 'Gizmo';
-- without WHERE, UPDATE or DELETE affects every row
```

The condition identifies the write target; Gizmo changes while the other prices remain

Caveat: A Write Without WHERE

```
UPDATE Product SET price=24.99;
```

↑ no WHERE: every row changes

```
SELECT pname,price FROM Product;
```

pname	price
Gizmo	24.99
PowerGizmo	24.99
SingleTouch	24.99
MultiTouch	24.99

All four prices change when UPDATE has no target condition

Caveat: Deleting a Referenced Company

```
DELETE FROM Company WHERE cname='GizmoWorks';
```

SQLITE SAYS

```
FOREIGN KEY constraint failed
```

A company referenced by products cannot be deleted under the default policy

Transactions

Acct	Balance
a10	20,000
a20	15,000

Transfer $3k from a10 to a20:

1. Debit $3k from a10
2. Credit $3k to a20

Acct	Balance
a10	17,000
a20	18,000

A transaction commits related changes together; ROLLBACK cancels uncommitted changes

Both balance changes must complete together; one completed change would lose money

ACID Describes Transaction Guarantees

Property	Meaning
Atomicity	Related changes succeed or fail together
Consistency	Declared constraints hold after the transaction
Isolation	Concurrent transactions do not expose incomplete work
Durability	Committed work survives a failure

ACID connects the write boundary to constraints, concurrency and recovery

Caveat: A Crash Between Two Writes

- A key concept is the **transaction (TXN)**: an **atomic** sequence of db actions (reads/writes)

Atomicity: An action either completes *entirely* or *not at all*

Acct	Balance
a10	20,000
a20	15,000

Transfer $3k from a10 to a20:

1. Debit $3k from a10
2. Credit $3k to a20

Acct	Balance
a10	17,000
a20	18,000

Written naively, in which states is **atomicity** preserved?

- Crash before 1,
- After 1 but before 2,
- After 2.

BEGIN, COMMIT and ROLLBACK

```
BEGIN;
UPDATE Product SET price=24.99 WHERE pname='Gizmo';
SELECT pname,price FROM Product WHERE pname='Gizmo';
ROLLBACK;
SELECT pname,price FROM Product WHERE pname='Gizmo';
```

undo everything since BEGIN

Recap: Valid Writes

cname
Canon
GizmoWorks
Hitachi

pname	price	manufacturer
Gizmo	19.99	GizmoWorks
PowerGizmo	29.99	GizmoWorks
SingleTouch	149.99	Canon
MultiTouch	203.99	Hitachi

Gizmo starts at 19.99; Google is absent from Company

Demo: Change a Price Safely

1. Change Gizmo to 24.99 inside a transaction, inspect it, then roll back and inspect it again
2. Try inserting a product made by Google, which is absent from Company

```
-- write your query here, then Run
```

QUERY RESULT

Write a query on the left, then Run.

Answer: Roll Back a Price Change

```
BEGIN;
UPDATE Product SET price=24.99 WHERE pname='Gizmo';
SELECT price FROM Product WHERE pname='Gizmo';
ROLLBACK;
SELECT price FROM Product WHERE pname='Gizmo';
```

price
19.99

ROLLBACK restores Gizmo to 19.99 in the current transaction

Answer: A Missing Manufacturer

```
INSERT INTO Product
VALUES('NewGizmo',10,'Gadgets','Google');
```

SQLITE SAYS

```
FOREIGN KEY constraint failed
```

The foreign key rejects a manufacturer absent from Company

Takeaway: SQL I

1. **SELECT-FROM-WHERE** chooses the rows and columns of one table
2. **DISTINCT, ORDER BY, LIMIT and IS NULL** make the result match the question
3. **Keys and transactions** keep writes valid and related changes together